

Heidi H. Gui

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EDUCATION:

Bachelors of Science in Physics, Minor in Studio Art

August 2023-present

The Ohio State University (*Anticipated Graduation Spring 2027*)

GPA: 3.85

Awards: Dean's List AU23, SP24, AU24, SP25, AU25, SP26 | Physics Summer Research Program (SRP) Scholarship (2024)

OBJECTIVE:

Seeking Ph.D. in physics to contribute to instrumentation and calibration of ultra high energy neutrino detectors, with experience spanning in-ice and air simulations, antenna theory, and data analysis.

RESEARCH EXPERIENCES:

Simulation of Antenna Response in RNO-G Experiment

Advisor: Prof. [Kaeli Hughes](#)

August 2025-present

- ★ Gauging detector performance of the Radio Neutrino Observatory in Greenland (RNO-G) with differing horizontally polarized (H-pol) antenna models
- ★ Using simulation in XFDTD to solve maxwells equations in ice, then extracting e-field and gain
- ★ Utilizing the NuRadioMC simulation package based in Python to simulate high energy neutrino events
- ★ Performed in depth literary review and 15 minute oral presentation on H-pol antennas to lab group
- ★ Writing an undergraduate research thesis to defend in Spring 27

Searching for Radio Background Sources in RNO-G Experiment

Advisor: Prof. Kaeli Hughes

July 2024-July 2025

- ★ Continued work from SRP under Prof. Kaeli Hughes with RNO-G Collaboration
- ★ Analyzed RNO-G database for radio noise produced by airplanes in order to help reduce as much background
- ★ Utilized Python and Ohio SuperComputer to analyze data from RNO-G
- ★ Preliminary findings were presented at the Conference for Undergraduate Women and Gender Minorities in Physics (CU*iP) Poster Session at Purdue University
- ★ Products of research were discussed in the RNO-G collaboration paper titled "Radio emission from airplanes as observed with RNO-G."

PUBLICATIONS:

RNO-G Collaboration. "[Radio emission from airplanes as observed with RNO-G.](#)" *Journal of Instrumentation*, vol. 20, no. 11, p. P11015, 2025. doi:10.1088/1748-0221/20/11/P11015.

I am a part of the RNO-G Collaboration authors list.

POSTER PRESENTATIONS/TALKS:

Spring Undergraduate Research Festival

April 2026

Ohio State University, [*Simulating RNO-G's Horizontally Polarized Antenna Response to Radio Emission from Neutrino Ice Shower*](#)

CU*iP Poster Session

January 2025

Purdue University, [*Detecting Airplanes through Analysis of Airplane Background Noise*](#)

SRP Poster Session

July 2024

Ohio State University, *Detecting Airplanes through Analysis of Airplane Background Noise*

Polaris Mentorship Program Poster Session

April 2024

Ohio State University, [*Constraining the Electron Neutrino Mass using SN1987A*](#)

The Undergraduate Residential Summer Access Program Poster Session

August 2023

Ohio State University, [*Dark Matter and Photon Interactions*](#)

CONFERENCES:

CCAPP's 20th Anniversary Conference

June 2026

Ohio State University

Conference for Undergraduate Women and Gender Minorities in Physics (CU*iP)

January 2025

Purdue University

Conference for Undergraduate Women in Physics (CUWiP)

January 2024

University of West Virginia

PROGRAMS:

Momentum Mentorship Program

August 2025-present

- ★ Mentoring an underclassman enrolled in the Intermediate Mechanics I and II sequence at OSU
- ★ Supporting mentee by meeting once every two weeks to discuss goals, study tips, imposter syndrome, and feelings about the course

AMGEN STEM Learning Center

January 2026-May 2026

- ★ Tutored students in a group setting for introductory level Physics 1 and Physics 2
- ★ Planned group tutoring sessions and grading practice problems
- ★ Led tutoring for groups of 6-8 taking introductory level Physics 1 and Physics 2 by meeting weekly to practice physics problems for 80 minutes, resulting in a better understanding of the physics.

Physics Summer Research Program (SRP)

May 2024-July 2024

- ★ Summer funded research through a stipend from tOSU Department of Physics
- ★ Worked under Prof. Kaeli Hughes to search for radio airplane background noise in RNO-G
- ★ Analyzed data with Python and presented a poster on the work at the end of the program

Polaris Mentorship Course

August 2023-May 2024

- ★ Enrolled in a mentorship course and was mentored by a graduate student in the physics department at OSU
- ★ Met with mentor twice a week to discuss professional development and topics in astroparticle physics
- ★ Conducted and presented a poster for a mini research project on constraining the electron-neutrino mass with mentor

The Undergraduate Residential Summer Access Program (URSA)

August 2023

- ★ Participated in early arrival two week program for underrepresented students in physics and astronomy/astrophysics
- ★ Learned and conducted a research mini project about dark matter with a group of peers
- ★ Presented poster to graduate students and faculty members of the physics and astrophysics/astronomy departments

VOLUNTEERING:

Achieving in Science through Physics Instrumentation Research and Exploration (ASPIRE) Workshop

June 2025, June 2026

- ★ Participated in undergraduate student panel to answer questions about college career
- ★ Helped advertise the program to high school students
- ★ Assisted students during the arduino and speed of sound project

Phun with Physics

July 2024

- ★ Helped facilitate the Physics departments science booth at the 2024 Ohio State Fair
- ★ Taught children elementary physics topics like sound and light waves

RELEVANT COURSEWORK:

Physics: Introductory Physics I and II; Intermediate Mechanics I and II; Quantum Mechanics I; Honors Electromagnetism I; Honors Electromagnetism II; Topics in Computational Physics, Theoretical Mechanics, Advanced Physics Laboratory

Math: Calculus I, II and III; Differential Equations; Linear Algebra

SKILLS:

Coding: Python, C++, Mathematica, HTML, Bash, SQL (SQLite)

Software: VSCode, NuRadioMC, XFtd, Mattak, ROOT, Jupyter, GitHub, SLURM/HPC, LabPlot

Languages: English (Native), Mandarin Chinese (Conversational), French (Elementary)